

Advances in Model Merging for Physics-Informed Neural Networks (PINNs)

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Abstract. Physics-Informed Neural Networks (PINNs) have emerged as a powerful paradigm for solving partial differential equations by incorporating physical laws directly into neural network training. However, scaling PINNs to large domains and capturing high-frequency solution components remains challenging due to increased complexity and optimization difficulties. In this work, we introduce a novel nonlinear model merging approach for PINNs that extends beyond traditional linear combinations of model parameters. We demonstrate our approach on the linear damped oscillator problem. The experimental results show consistent improvements over both pretrained baselines and linear merging approaches, suggesting that the novel technique can effectively assist not only PINNs on larger domains but also facilitate integration of specialized submodels for large-scale foundation models.

Keywords: Physics-Informed Neural Networks, Computational Mechanics, Structural Dynamics, Model Merging

1 Introduction

Over the past decades, machine learning has experienced remarkable advancements, progressing from simple linear models and decision trees to sophisticated deep learning architectures capable of addressing complex tasks in vision, language, and decision-making LeCun et al. [1]. The increasing availability of large datasets and computational resources has driven breakthroughs in areas such as image recognition Krizhevsky et al. [2], natural language processing Devlin et al. [3], and reinforcement learning Mnih et al. [4], enabling machines to achieve or even surpass human-level performance in certain domains.

Among these advancements, model merging has emerged as an effective technique in the machine learning community, enabling the combination of parameters of multiple trained models into a single model without requiring access to raw training data or significant computational resources Yang et al. [5]. This approach enables the integration of complementary skills, knowledge, or specializations learned by different models, and has shown promise in scenarios such as continual learning, federated learning, and multitask adaptation Yang et al. [6]. As its application expands across diverse domains, it becomes increasingly important to understand model merging methods.

In parallel with these developments, researchers have been exploring machine learning's potential beyond traditional applications, seeking to address challenges in scientific and engineering disciplines. This gave rise to the field of *scientific machine learning* (SciML), which integrates the strengths of machine learning to enhance, accelerate, or even automate the discovery of physical laws, the solution of differential equations, and the modeling of complex systems Hao et al. [7], Karniadakis et al. [8].

SciML techniques have demonstrated remarkable success in areas such as Physics-Informed Neural Networks (PINNs), a method that integrates neural networks with physical laws to solve differential equations by training on data that satisfies the governing equations of the physical system, thereby enabling the models to learn the underlying physics and make accurate predictions about system behavior Raissi et al. [9].

However, significant challenges remain for PINNs, particularly when scaling to large problem domains and capturing high-frequency solution components. As the domain grows, solutions become more complex, requiring larger networks and making optimization harder and slower. Thus, most successful PINN applications have been restricted to relatively small domains and problems with limited frequency content, while multi-scale and large-domain problems remain a major open challenge Moseley et al. [10].

In this work, inspired by the concept of model editing using task vectors introduced by Ilharco et al. [11], we introduce a *nonlinear* model merging scheme for PINNs. Our goals are to assist PINNs on larger domains, and to

empirically assess and extend task-vector techniques, demonstrating consistent improvements over pretrained and linear-merge baselines.

2 Methodology

Problem Definition. Let a general system of partial differential equations (PDEs) be defined in a domain $[\Omega \subset \mathbb{R}^d]$:

$$F[\mathbf{y}(x)] = 0, \quad x = (x^1, x^2, \dots, x^d) \in \Omega, \quad (1)$$

$$\mathbf{y} = (y_1(x), y_2(x), \dots, y_n(x)) \in \mathbb{R}^n, \quad (2)$$

with boundary conditions specified by:

$$B[\mathbf{y}(x)] = 0, \quad x \in \partial\Omega. \quad (3)$$

The fundamental idea of PINNs is to approximate $\mathbf{y}(x)$ using a multi-layer feed-forward neural network with $N_\theta(x)$ layers, parameterized by weights and biases collected in θ , i.e.,

$$\mathbf{y}(x) \approx N_\theta(x). \quad (4)$$

The training of the hyperparameters θ that best satisfy the governing equations is performed by minimizing the residual resulting from not exactly fulfilling these equations. In the context of structural dynamics, based on the problem's information, the PINN optimizes a loss function that considers the structural equilibrium equations and boundary conditions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{net}} + \mathcal{L}_{\text{constrained}}. \quad (5)$$

During the training process, the neural network adjusts its parameters to minimize this loss function. This approach provides an efficient alternative for solving structural analysis problems, as it eliminates the need to explicitly formulate the global stiffness matrix, making it particularly useful for complex or nonlinear problems.

Proposed Method. Let $\theta_{\text{pre}} \in \mathbb{R}^d$ be the parameters of a pre-trained base PINN, and let $\theta_{ft}^t \in \mathbb{R}^d$ be the parameters after fine-tuning on subdomain t . Following the approach of Ilharco et al. [11], we propose to merge multiple specialized PINNs — each trained on a different subdomain or task — by **nonlinearly** combining their parameter updates, weighting each with scaling coefficients λ_k, λ_N :

$$\theta_M = \theta_{\text{pre}} + \sum_{k=1}^K \lambda_k \left(\sum_{N=1}^{\text{Order}} \lambda_N \left(\theta_{ft}^{t_k} - \theta_{\text{pre}} \right)^N \right). \quad (6)$$

Experiments. We solve the linear damped oscillator

$$u''(t) + 2d u'(t) + \omega_0^2 u(t) = 0, \quad t \in [0, 1], \quad (7)$$

with initial conditions $u(0) = 1$ and $u'(0) = 0$. We set $d = 2$ and $\omega_0 = 20$ (so $\mu = 2d$ and $k = \omega_0^2$). The exact solution is available in closed form and is used for evaluation.

Training protocol. We first pretrain a single PINN on the whole domain using a physics loss on the collocation points plus the initial conditions. Next, we split the time interval into $K = 10$ subdomains and fine-tune one copy of the network per subdomain, enforcing the ODE residual inside each subdomain, the boundary conditions at $t = 0$ and $t = 1$, and C^0/C^1 continuity at all internal interfaces.

Model merging. After fine-tuning, we merge parameters by expanding per-parameter deltas $\Delta\theta_k = \theta_{ft}^{t_k} - \theta_{\text{pre}}$ up to a chosen *Order* (see Figures 1 and 2). In our implementation, $\Delta\theta_k$ is taken element-wise and we consider two families of scaling coefficients:

- **Taylor:** one small Multi-Layer Perceptron (MLP) produces, per parameter, model-wise weights λ_k , and λ_N coefficients are taken as the fixed Taylor coefficients $1/N!$ up to the chosen order.
- **Learned 2 MLPs:** two small MLPs produce, per parameter, (i) order-wise weights λ_N and (ii) model-wise weights λ_k ; both are softmax-normalized and trained jointly to minimize validation Mean Squared Error (MSE).

3 Results and Discussion

The results demonstrate the effectiveness of our proposed nonlinear model merging approach compared to the linear approach of Ilharco et al. [11]. Figures 1 and 2 show that the merged models achieve excellent convergence to the exact solution across different polynomial orders. Even for low orders (e.g., 1–3), the models significantly outperform the pre-trained baseline, capturing both the amplitude and phase of the oscillatory dynamics with high fidelity. As the order increases, phase lag and amplitude errors are progressively reduced, with orders 5–7 producing curves that are visually indistinguishable from the exact solution.

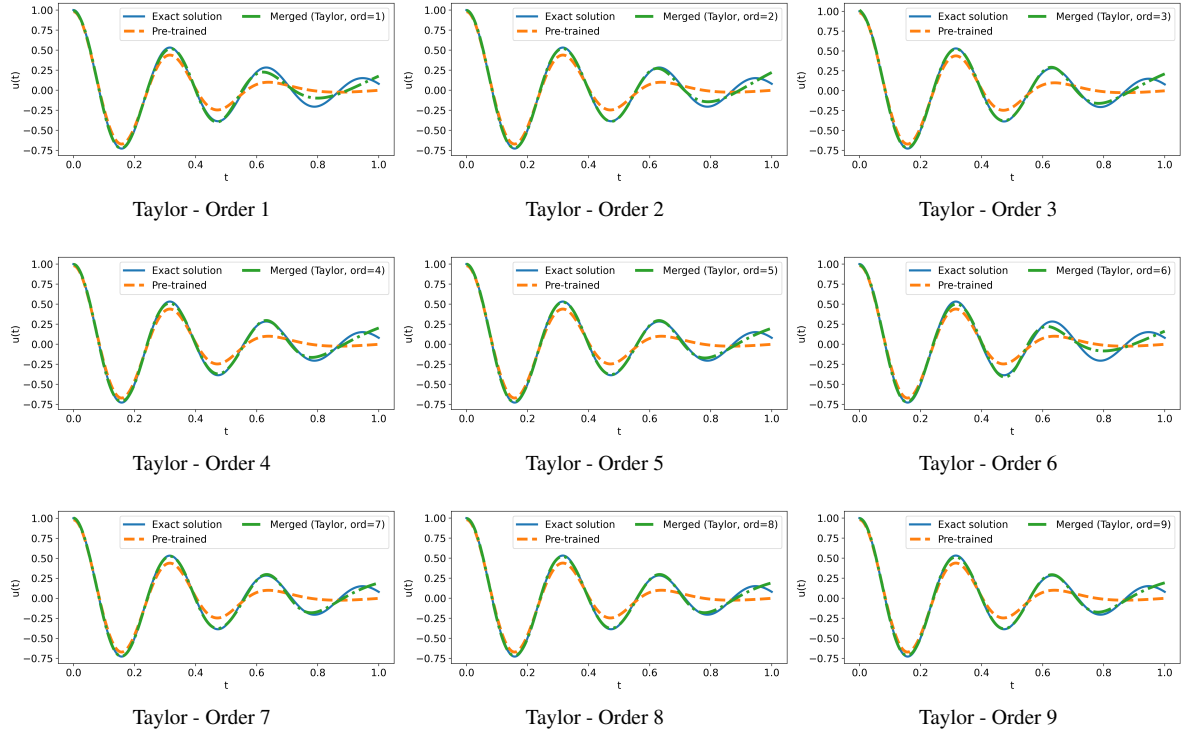


Figure 1. Taylor model merging results for orders 1-9.

Order	Pre-Trained (%)	2 MLPs (%)	Taylor (%)
1	151.56	63.97	63.97
2	151.56	36.53	39.06
3	151.56	50.19	62.55
4	151.56	22.48	58.55
5	151.56	12.70	53.52
6	151.56	12.22	51.00
7	151.56	12.01	54.30
8	151.56	21.09	55.32
9	151.56	58.57	68.95

Table 1. Comparison of error rates for the models by order.

This trend is corroborated by Figure 3, which summarizes the performance of the different merging strategies. Both merging strategies massively outperform the pre-trained baseline and the linear approach, with the learned merging using two small MLPs consistently achieving the lowest relative mean error – reaching an optimal range for orders 5-7. Very low orders underfit, while very high orders (≥ 8) begin to degrade, most notably order 9 for the learned method. We attribute this high-order degradation to:

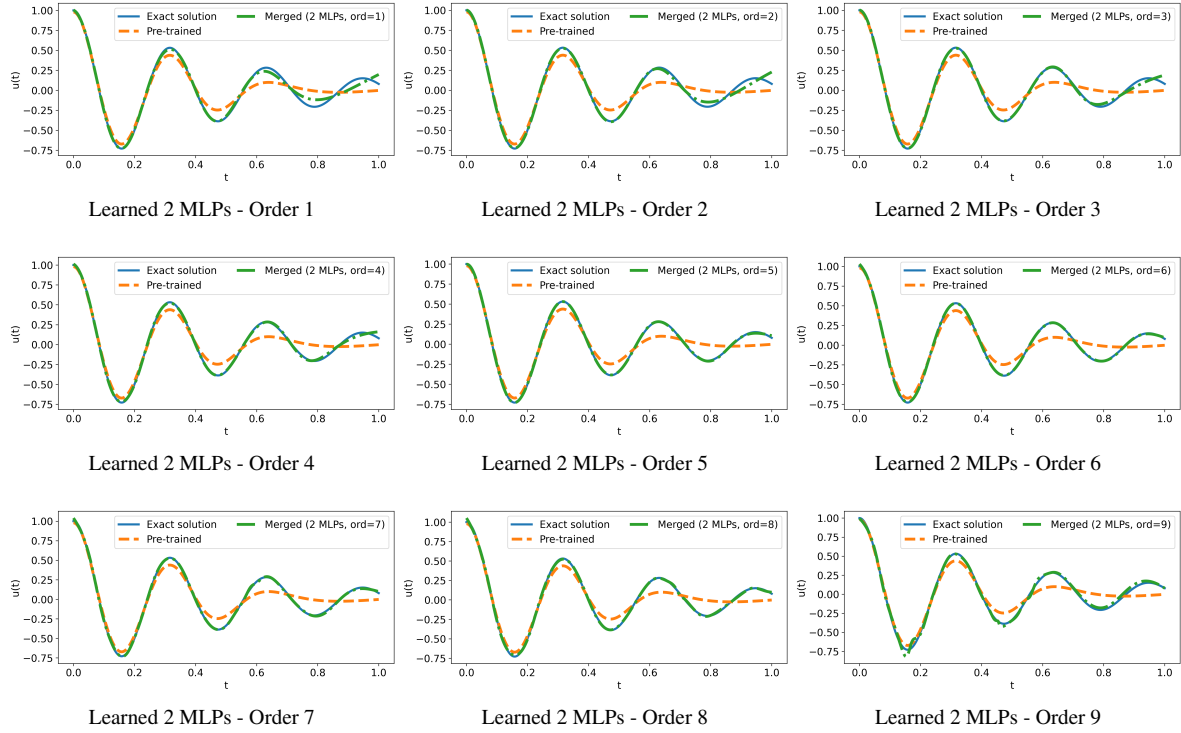


Figure 2. Learned 2 MLPs model merging results for orders 1-9.

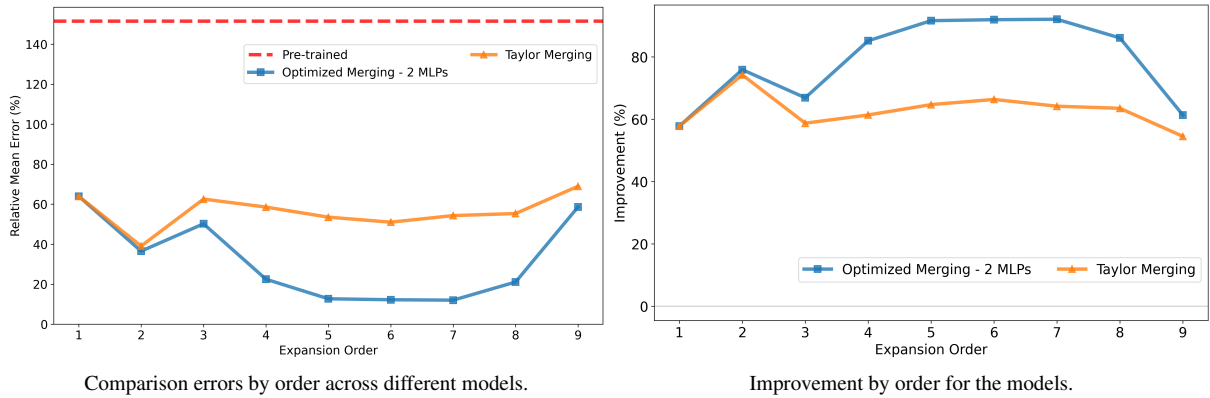


Figure 3. Performance comparison and improvement analysis by order.

- **Ill-conditioning of powers of the update:** $(\Delta\theta)^N$ magnifies outliers and drives small entries toward machine zero, producing poorly scaled merged weights;
- **Local-approximation breakdown:** the expansion is only valid near θ_{pre} , so high-order partial sums fail to approximate accurately when fine-tuned solutions lie farther away;
- **Layer/scale mismatch:** differences in magnitude and sensitivity between layers are amplified by high-order terms, overshadowing beneficial lower-order structure.

Additionally, from Table 1 and Figure 3, it is evident that the **Learned** approach yields an improvement of approximately 85% – 92% for orders 4-7, peaking around orders 5-7, whereas the **Taylor** merging method plateaus at roughly 58% – 66%. This performance gap underscores that learning both the model weights and the per-order coefficients enables the capture of non-linear corrections that fixed Taylor coefficients cannot. The $1/N!$ schedule implicitly assumes a true Taylor expansion in weight space – an assumption that the merge map generally does not satisfy – limiting its capacity to adapt to the complex, non-linear structure of the fine-tuned parameter updates.

Finally, it is possible to visualize in Figure 4 a direct comparison between fine-tuned PINNs trained independently on individual subdomains and the merged global solution. The merged model closely matches the fine-tuned predictions within each subdomain while maintaining continuity across interfaces, highlighting the effectiveness

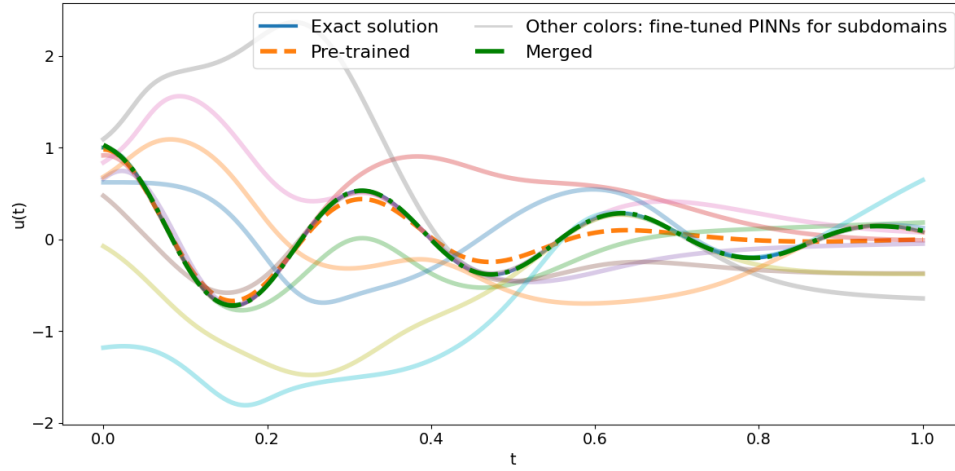


Figure 4. Visual representation of the fine-tuned PINNs for the subdomains.

of the merging strategy in synthesizing localized expertise into a coherent global model.

4 Conclusions

In this work, we introduced a novel polynomial *task vector* merging approach for Physics-Informed Neural Networks (PINNs) that fundamentally extends beyond traditional linear parameter combinations. By incorporating nonlinear terms up to a specified polynomial order, our method captures significantly richer interactions between specialized model parameters trained on different subdomains or tasks. This approach leverages the mathematical structure of power series expansion in parameter space to systematically combine the learned adaptations from multiple fine-tuned networks, enabling more sophisticated knowledge integration than simple weighted averaging schemes.

Our experimental evaluation on the linear damped oscillator problem demonstrates that nonlinear merging substantially outperforms both linear merging approaches and pretrained baseline models. The findings suggest that polynomial task vectors with learned weights offer a computationally efficient and theoretically grounded alternative for merging specialized neural network models. Unlike traditional ensemble methods that require maintaining multiple full models, our approach produces a single merged model that inherits the specialized knowledge from each subdomain-specific network while maintaining global coherence and continuity constraints.

The practical implications extend well beyond PINNs to broader machine learning applications. Such nonlinear merging techniques could significantly benefit large-scale foundation models in natural language processing and the recently released *PhysiX* framework for physics simulations. These methods enable efficient integration of specialized submodels or domain-adapted experts without the computational costly retraining from scratch, making them particularly valuable for scaling neural networks to complex, multi-domain problems.

Future work should explore the theoretical foundations of polynomial task vector spaces, investigate optimal order selection strategies, and extend the approach to more complex PDE systems and higher-dimensional domains. Additionally, developing regularization techniques to mitigate high-order degradation could further enhance the robustness and applicability of this merging paradigm.

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